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METHOD FOR MANUFACTURING ELECTRONIC COMPONENT DEVICE

Abstract

A method for manufacturing an electronic device, which includes, preparing an intermediate structure including a base material and a wiring layer, the base material having a first main surface and a second main surface on a rear side of the first main surface, the wiring layer being provided on the first main surface and having an insulating resin layer and wiring, the base material having a resin portion including a through portion penetrating from the first main surface to the second main surface, the wiring layer forming a trench having a bottom surface on which the through portion is exposed; and cutting the through portion along the trench, thereby forming a plurality of wiring structures, each having the divided base material.

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Background/Summary

TECHNICAL FIELD
[0001] The present disclosure relates to a method for manufacturing an electronic component device.
BACKGROUND ART
[0002] An exemplary semiconductor package with a plurality of semiconductor components arranged two-dimensionally is a so-called 2.3-dimensional type, which features an interposer with fine wiring for connecting the multiple semiconductor components (e.g., Patent Literatures 1 and 2).
CITATION LIST
Patent Literature
[0003] Patent Literature 1: U.S. Patent Application Publication No. 2021/0074646.
[0004] Patent Literature 2: U.S. Patent Application Publication No. 2021/0074645.
SUMMARY OF INVENTION
Technical Problem
[0005] In a method for manufacturing an electronic component device that has an insulating resin layer and a wiring layer including wiring, multiple wiring structures may be formed from a single intermediate structure on which a wiring layer is formed by cutting a base material, which includes a resin portion, along with the wiring layer. However, since it is typical that the insulating resin layer constituting the wiring layer and the resin portion constituting the base material have different physical properties like hardness, cutting both at the same time using the same method is likely to cause a defect such as damage to one of them. This is particularly challenging when the base material includes a relatively hard resin portion that contains a lot of inorganic filler, as a cutting method suitable for cutting the resin portion is liable to cause delamination or damage to the insulating resin layer that constitutes the wiring layer.
Solution to Problem
[0006] The present disclosure includes the following items.
[0007] [1] A method for manufacturing an electronic component device, comprising:
[0008] preparing an intermediate structure comprising a base material and a wiring layer, the base material having a first main surface and a second main surface on a rear side of the first main surface, the wiring layer being provided on the first main surface and comprising an insulating resin layer and wiring provided in the insulating resin layer, the base material comprising a resin portion comprising a through portion penetrating from the first main surface to the second main surface, the wiring layer forming a trench having a bottom surface on which the through portion is exposed; and
[0009] cutting the through portion along the trench, thereby forming a plurality of wiring structures, each comprising the divided base material and the wiring layer provided on the base material.
[0010] [2] The method according to [1], in which
[0011] the resin portion comprises an inorganic filler,

[0012] the insulating resin layer does not comprise an inorganic filler, or the insulating resin layer comprises the inorganic filler, and
[0013] in a case where the insulating resin layer comprises the inorganic filler, a ratio of a volume of the inorganic filler comprised in the insulating resin layer to a volume of the insulating resin layer is smaller than a ratio of a volume of the inorganic filler comprised in the resin portion to a volume of the resin layer.
[0014] [3] The method according to [1], in which the intermediate structure is prepared in a way that includes forming the wiring layer forming the trench on the base material.
[0015] [4] The method according to [1], in which the intermediate structure is prepared in a way that includes forming the wiring layer forming the trench on a carrier substrate, and shifting the wiring layer from the carrier substrate onto the base material.
[0016] [5] The method according to any one of [1] to [4], in which
[0017] the wiring layer is
[0018] formed in a way that includes repeating forming a pattern layer through exposure and development of a photosensitive resin layer and forming a conductor layer, the pattern layer having a pattern including an opening for the wiring and an opening for the trench, the conductor layer comprising a via portion that fills the opening for the wiring, and
[0019] the insulating resin layer is formed by a plurality of the pattern layers, the wiring is formed by a plurality of the conductor layers, and the trench is formed by connecting a plurality of the openings for the trench formed by a plurality of the pattern layers.
[0020] [6] The method according to any one of [1] to [4], in which
[0021] the wiring layer is
[0022] formed in a way that includes repeating removing a part of a resin layer with laser irradiation to form a pattern layer and forming a conductor layer, the pattern layer having a pattern including an opening for the wiring and an opening for the trench, the conductor layer comprising a via portion that fills the opening for the wiring, and
[0023] the insulating resin layer is formed by a plurality of the pattern layers, the wiring is formed by a plurality of the conductor layers, and the trench is formed by connecting a plurality of the openings for the trench formed by a plurality of the pattern layers.
[0024] [7] The method according to any one of [1] to [4], in which
[0025] the wiring layer is
[0026] formed in a way that includes repeating forming a pattern layer through exposure and development of a photosensitive resin layer and forming a conductor layer, the pattern layer having a pattern including an opening for the wiring, the conductor layer comprising a via portion that fills the opening for the wiring, and
[0027] the insulating resin layer is formed by a plurality of the pattern layers, the wiring is formed by a plurality of the conductor layers, and a part of the formed insulating resin layer is removed with laser irradiation to form the trench.
[0028] [8] The method according to any one of [1] to [7], further comprising mounting a plurality of semiconductor components on the wiring layer,
[0029] in which each of the plurality of wiring structures being formed has one or more of the semiconductor components.
[0030] [9] The method according to any one of [1] to [7], in which
[0031] the base material further comprises an internal wiring layer exposed on the second main surface,
[0032] the internal wiring layer forms an internal trench filled with the through portion of the resin portion,
[0033] the method further comprises mounting a plurality of semiconductor components on the internal wiring layer,
[0034] the through portion is cut along the trench and the internal trench, and

[0035] each of the plurality of wiring structures being formed comprises one or more of the semiconductor components.
[0036] [10] The method according to [8] or [9], in which
[0037] the base material further comprises a relay wiring portion comprising relay wiring electrically connected to the plurality of semiconductor components, the relay wiring portion being sealed by the resin portion, and
[0038] each of the plurality of wiring structures being formed comprises the relay wiring portion and two or more of the semiconductor components electrically connected via the relay wiring portion.
[0039] [11] The method according to any one of [1] to [10], in which the width of the trench increases in a direction away from the base material.
[0040] [ ] The method according to any one of [1] to [11], further comprising mounting the wiring structure on an organic wiring substrate.
Advantageous Effects of Invention
[0041] A method is disclosed for easily manufacturing a wiring structure including a base material that has a resin portion containing a lot of inorganic fillers and a wiring layer provided on the base material. This method is applicable, for example, to the fabrication of 2.3-dimensional semiconductor packages.

Description

BRIEF DESCRIPTION OF DRAWINGS
[0042] FIG. **1** is a process diagram illustrating an example of a method for manufacturing an electronic component device.
[0043] FIG. **2** is a process diagram illustrating an example of a method for manufacturing an electronic component device.
[0044] FIG. **3** is a process diagram illustrating an example of a method for manufacturing an electronic component device.
[0045] FIG. **4** is a process diagram illustrating an example of a method for manufacturing an electronic component device.
[0046] FIG. **5** is a process diagram illustrating an example of a method for manufacturing an electronic component device having a semiconductor component.
[0047] FIG. **6** is a process diagram illustrating an example of a method for manufacturing an electronic component device having a semiconductor component.
[0048] FIG. **7** is a process diagram illustrating an example of a method for manufacturing an electronic component device having a semiconductor component.
[0049] FIG. **8** is a process diagram illustrating an example of a method for manufacturing an electronic component device having a semiconductor component.
[0050] FIG. **9** is a process diagram illustrating an example of a method for manufacturing an electronic component device having a semiconductor component.
[0051] FIG. **10** is a process diagram illustrating an example of a method for manufacturing an electronic component device having a semiconductor component.
[0052] FIG. **11** is a process diagram illustrating an example of a method for manufacturing an electronic component device having a semiconductor component.
DESCRIPTION OF EMBODIMENTS
[0053] The present disclosure is not limited to the following examples. In the following examples, duplicated descriptions may be omitted.
[0054] FIGS. **1** and **2** are process diagrams illustrating an example of a method for manufacturing an electronic component device. The method illustrated in FIG. **1** and FIG. **2** includes preparing an

intermediate structure **10**A having a base material **1** and a wiring layer **7**. The base material **1** has a first main surface S**1** and a second main surface S**2** on the rear side of the first main surface S**1**. The wiring layer **7** is provided on the first main surface S**1** and has an insulating resin layer **3** and wiring **5** provided in the insulating resin layer **3**. The base material **1** has a resin portion **20** including a through portion **20**A penetrating from the first main surface S**1** to the second main surface S**2**. The insulating resin layer **3** forms a trench T having a bottom surface that exposes the through portion **20**A. The method also includes cutting the through portion **20**A along the trench T to form a plurality of wiring structures **10**, each having the base material **1** divided and the wiring layer **7** provided on the base material **1**.

[0055] The through portion **20**A is a portion of the resin portion **20** that penetrates from the first main surface SI to the second main surface S**2**. The resin portion **20** includes at least the through portion **20**A and can be a member integrally formed from a resin material such as a sealing material. The entire first main surface S**1** of the base material **1** may also be the surface of the resin portion **20**. Alternatively, a relay wiring portion, which will be described later, may be provided in the base material **1** separately from the resin portion **20** including the through portion **20**A, and the relay wiring portion may be exposed on the first main surface S**1**.

[0056] The resin portion **20** includes a resin and an inorganic filler. The insulating resin layer **3** may contain a resin and an inorganic filler. In the case where the insulating resin layer **3** contains an inorganic filler, the ratio of the volume of the inorganic filler contained in the insulating resin layer **3** to the volume of the insulating resin layer **3** is smaller than the ratio of the volume of the inorganic filler contained in the resin portion **20** to the volume of the resin portion **20**. Due to the difference in the ratio of inorganic fillers, the resin portion **20** is relatively harder. Different hardnesses often result in different suitable cutting conditions, but according to the method disclosed herein, since only the resin portion **20** (the through portion **20**A) is cut, it is possible to employ cutting conditions suitable for cutting the resin portion **20** while avoiding any influence on the insulating resin layer **3**. The resin portion **20** (the through portion **20**A) is cut, for example, by a rotating blade.

[0057] The ratio of the volume of the inorganic filler in the resin portion **20** to the volume of the resin portion **20** may be, for example, 10% or more and 95% or less by volume. The ratio of the volume of the inorganic filler in the insulating resin layer **3** to the volume of the insulating resin layer **3** may be, for example, 0% or more and 50% or less by volume.

[0058] In the example of FIGS. **1** and **2**, the wiring layer **7** is formed by repeatedly forming a pattern layer **3***a* having a pattern including an opening for the wiring **35** and an opening for the trench **37** by exposing and developing a photosensitive resin layer **30** formed on the first main surface S**1** of the base material **1**, and forming a conductor layer **5***a* including a via portion **51** filling the opening for the wiring **35** and a wiring pattern portion **52** provided on the pattern layer **3***a*. The insulating resin layer **3** is formed by the first pattern layer **3***a*, a second pattern layer **3***b*, and a third pattern layer **3***c*, which are formed in sequence from the side of the base material **1**. The wiring **5** is formed by the first conductor layer **5***a*, a second conductor layer **5***b*, and a third conductor layer **5***c*, which are formed in sequence from the side of the base material **1**. The trench T penetrating the insulating resin layer **3** is formed by connecting the multiple openings for the trench **37** formed by the plurality of pattern layers **3***a*, **3***b*, and **3***c*.

[0059] The photosensitive resin layer **30** and the pattern layers **3***a*, **3***b*, and **3***c* can be formed using conventional resist materials that are used to form an insulating resin layer of a wiring layer. For the exposure of the resin layer **30**, active light rays such as ultraviolet light are applied through a mask **9**, which has an opening provided at a position corresponding to the opening for the wiring **35** and the opening for the trench **37**. Instead of using the exposure and development method, a part of the resin layer **30** may be removed with laser irradiation to form the pattern layers **3***a*, **3***b*, and **3***c*, which have a pattern including the opening for the wiring **35** and the opening for the trench **37**. In this case, the resin layer **30** may be non-photosensitive.

[0060] The conductor layers **5***a*, **5***b*, and **5***c* and the wiring **5** can be formed using conventional methods such as plating, printing of conductor paste, or sputtering.
[0061] The wiring layer **7** is used, for example, as a redistribution layer connected to a semiconductor component including an IC chip. The number of pattern layers and conductor layers that constitute the wiring layer **7** is not limited to a particular number, but may be, for example, **2** or more and 8 or less, respectively. The thickness of the entire wiring layer **7** may be, for example, 10 μm or more and 150 μm or less.
[0062] The intermediate structure **10**A may be prepared by a method that includes forming the wiring layer **7**, which forms the trench **1**, on a carrier substrate separate from the base material **1**, and shifting the wiring layer **7** from the carrier substrate onto the base material **1**.
[0063] FIGS. **3** and **4** are process diagrams illustrating another exemplary method for manufacturing an electronic component device. In the case of the method illustrated in FIGS. **3** and **4**, the wiring layer **7** is formed in a way, which includes repeatedly forming a pattern layer **3***a* having a pattern including an opening for the wiring **35** by exposure and development of the photosensitive resin layer **30** and forming a conductor layer **5***a* including a via portion **51** that fills the opening for the wiring **35** and a wiring pattern portion **52** provided on the pattern layer **3***a*. As in the method for FIGS. **1** and **2**, the insulating resin layer **3** is formed by the plurality of pattern layers **3***a*, **3***b*, and **3***c*, and the wiring **5** is formed by the plurality of conductor layers **5***a*, **5***b*, and **5***c*. Then, as illustrated in FIG. **4**(*f*), a part of the formed insulating resin layer **3** is removed with laser irradiation to form the trench T.
[0064] As illustrated in another example in FIG. **5**, the trench T may be formed with a width that widens in the direction away from the base material **1**. By varying the width of the opening for the trench formed by the multiple pattern layers **3***a*, **3***b*, and **3***c*, it is possible to form the trench T with a gradually widening width. In the case where the end face of the trench T (insulating resin layer **3**) is inclined in this way, the occurrence of cracks or delamination starting from the edge of the insulating resin layer **3** can be suppressed.
[0065] FIGS. **6**, **7**, **8**, and **9** are process diagrams illustrating an example of a method for manufacturing an electronic component device having multiple semiconductor components. In the method illustrated in FIGS. **6** to **9**, as illustrated in FIG. **6**, the base material **1** is formed in a way that includes forming an internal wiring layer **7**A forming an internal trench Ta, arranging a relay wiring portion **4** and a copper pillar **8** on the internal wiring layer **7**A, forming the resin portion **20** that seals the relay wiring portion **4**, and removing the surface layer portion of the resin portion **20** opposite the internal wiring layer **7**A to form a flat surface that exposes the relay wiring portion **4** and the copper pillar **8**. The base material **1** being formed includes the internal wiring layer **7**A, the relay wiring portion **4**, and the resin portion **20**. The resin portion **20** includes a portion that fills the internal trench Ta of the internal wiring layer **7**A and includes a through portion **20**A that penetrates from the first main surface S**1** to the second main surface S**2** on the rear side. The internal wiring layer **7**A is exposed on the second main surface S**2** of the base material **1**. The internal wiring layer **7**A can be formed by exposure and development of a photosensitive resin layer, laser irradiation, or a combination of these, similar to the method illustrated in FIGS. **1** to **4**.
[0066] The relay wiring portion **4** has a main body **41** including relay wiring electrically connected to a plurality of semiconductor components and has a terminal **42** provided on the outer surface of the main body **41**. The relay wiring portion **4** is arranged on the internal wiring layer **7**A in the orientation in which the terminal **42** is positioned on the side opposite the internal wiring layer **7**A. The relay wiring portion **4** may be a silicon interposer including a silicon substrate. The copper pillar **8** can be formed using conventional by conventional methods such as plating or printing of a conductive paste. The copper pillar **8** is electrically connected to the wiring **5** in the internal wiring layer **7**A.
[0067] The resin portion **20** can be formed using conventional sealing materials such as a thermosetting resin composition containing an inorganic filler. The inorganic filler may include, for

example, silica particles.
[0068] Subsequently, as illustrated in FIG. **7**(*e*), a wiring layer **7**B is formed on the first main surface SI of the base material **1**, forming a trench Tb having a bottom surface that exposes the through portion **20**A. The trench Tb is formed at a position overlapping with the internal trench Ta. The wiring layer **7**B can be formed using a method similar to that illustrated in FIGS. **1** to **4**. The wiring **5** in the wiring layer **7**B is electrically connected to the relay wiring portion **4** and the copper pillar **8**.
[0069] On the formed wiring layer **7**B, a plurality of semiconductor components **71** and **72** are mounted (FIG. **7**(*f*)). The semiconductor components **71** and **72** each have a bump **55**, and the semiconductor components **71** and **72** are electrically connected to the wiring layer **7**B via the bump **55**. The space between the semiconductor components **71** and **72** and the wiring layer **7**B is filled with an underfill material **25**.
[0070] The intermediate structure **10**A having the base material **1**, the wiring layer **7**B, and the semiconductor components **71** and **72** is shifted onto a carrier substrate **61**, which is separate from the carrier substrate **60**, in the orientation in which the semiconductor components **71** and **72** are positioned on the side of the carrier substrate **61**, and in this state, a bump **15** is provided on the internal wiring layer **7**A (FIG. **7**(*g*)).
[0071] Then, as illustrated in FIG. **8**(*a*), the intermediate structure **10**A is shifted onto another carrier substrate **62**. The carrier substrate **62** has a support substrate **62**A and a temporary fixing material layer **62**B provided on the support substrate **62**A. The intermediate structure **10**A is temporarily fixed to the carrier substrate **62** in the orientation in which the bump **15** contacts the temporary fixing material layer **62**B. In this state, the through portion **20**A of the resin portion **20** is cut from the side of the trench Tb along the trench Tb and the internal trench Ta, and thus a plurality of wiring structures **10** is formed on the carrier substrate **62** (FIG. **8**(*i*)).
[0072] The wiring structure **10** is delaminated from the carrier substrate **62** (FIG. **8**(*j*)). The wiring structure **10** is an electronic component device having the relay wiring portion **4** and the plurality of semiconductor components **71** and **72**, that is, a semiconductor package. The plurality of semiconductor components **71** and **72** are electrically connected through the relay wiring portion **4**. The semiconductor components **71** and **72** constituting one wiring structure **10** can be components having different functions. For example, the semiconductor component **71** may be a system-on-chip (SoC), and the semiconductor component **72** may be a memory. A single wiring structure **10** (semiconductor package) may also have a plurality of semiconductor components of the same type.
[0073] As illustrated in FIG. **9**, the wiring structure **10** is mounted on an organic wiring substrate **80** to obtain an electronic component device **100**. The wiring structure **10** is electrically connected to the organic wiring substrate **80** via the bump **15**. The underfill material **25** may be filled between the wiring structure **10** and the organic wiring substrate **80**. Various electronic components other than the wiring structure **10** may be mounted on one organic wiring substrate **80**.
[0074] FIGS. **10** and **11** are process diagrams illustrating another example of a method for manufacturing an electronic component device having multiple semiconductor components. The method illustrated in FIGS. **10** and **11** differs from the methods illustrated in FIGS. **7** to **9** in that the relay wiring portion **4** is arranged on the internal wiring layer **7**A in the orientation in which the terminal **42** is positioned on the side of the internal wiring layer **7**A (FIG. **10**(*b*)), and in that multiple semiconductor components **71** and **72** are mounted on the internal wiring layer **7**A (FIG. **11**(*f*)). The bump **15** is provided on the wiring layer **7**B with the intermediate structure **10**A temporarily fixed to the carrier substrate **60** (FIG. **11**(*e*)).
[0075] As illustrated in FIGS. **11**(*f*) and **11**(*g*), while the intermediate structure **10**A is temporarily fixed to the carrier substrate **62**, the through portion **20**A of the resin portion **20** is cut from the side of the internal trench Ta along the trench Tb and the internal trench Ta to form the plurality of wiring structures **10** on the carrier substrate **62**. The formed wiring structures **10** can be delaminated from the carrier substrate **62** and mounted on an organic wiring substrate.

Reference Signs List

[0076] **1** base material [0077] **3** insulating resin layer [0078] **5** wiring [0079] **3***a*, **3***b*, **3***c* pattern layer [0080] **4** relay wiring portion [0081] **5***a*, **5***b*, **5***c* conductor layer [0082] **7**, **7**B wiring layer [0083] **7**A internal wiring layer [0084] **10** wiring structure (semiconductor package) [0085] **10**A intermediate structure [0086] **20** resin portion [0087] **20**A through portion [0088] **30** resin layer [0089] **35** opening for wiring [0090] **37** opening for trench [0091] **51** via portion [0092] **52** wiring pattern portion [0093] **71**, **72** semiconductor component [0094] **80** organic wiring substrate [0095] **100** electronic component device [0096] S**1** first main surface of base material [0097] S**2** second main surface of base material [0098] T, Tb trench [0099] Ta internal trench

Claims

1. A method for manufacturing an electronic component device, comprising: preparing an intermediate structure comprising a base material and a wiring layer, the base material having a first main surface and a second main surface on a rear side of the first main surface, the wiring layer being provided on the first main surface and comprising an insulating resin layer and wiring provided in the insulating resin layer, the base material comprising a resin portion comprising a through portion penetrating from the first main surface to the second main surface, the wiring layer forming a trench having a bottom surface on which the through portion is exposed; and cutting the through portion along the trench, thereby forming a plurality of wiring structures, each comprising the divided base material and the wiring layer provided on the base material.

2. The method according to claim 1, wherein the resin portion comprises an inorganic filler, the insulating resin layer does not comprise an inorganic filler, or the insulating resin layer comprises the inorganic filler, and in a case where the insulating resin layer comprises the inorganic filler, a ratio of a volume of the inorganic filler comprised in the insulating resin layer to a volume of the insulating resin layer is smaller than a ratio of a volume of the inorganic filler comprised in the resin portion to a volume of the resin portion.

3. The method according to claim 1, wherein the intermediate structure is prepared in a way that includes forming the wiring layer forming the trench on the base material.

4. The method according to claim 1, wherein the intermediate structure is prepared in a way that includes forming the wiring layer forming the trench on a carrier substrate, and shifting the wiring layer from the carrier substrate onto the base material.

5. The method according to claim 1, further comprising: repeating a process of forming a pattern layer through exposure and development of a photosensitive resin layer to form a plurality of pattern layers, each of the pattern layers having a pattern including an opening for the wiring and an opening for the trench, the plurality of the pattern layers forming a plurality of openings for the trench; and repeating a process of forming a conductor layer comprising a via portion that fills the opening for the wiring to form a plurality of conductor layers, wherein the insulating resin layer of the wiring layer is formed by the plurality of the pattern layers, the wiring of the wiring layer is formed by the plurality of the conductor layers, and the trench is formed by connecting the plurality of the openings for the trench.

6. The method according to claim 1, further comprising: repeating a process of forming a pattern layer by removing a part of a resin layer with laser irradiation to form a plurality of pattern layers, each of the pattern layers having a pattern including an opening for the wiring and an opening for the trench, the plurality of the pattern layers forming a plurality of openings for the trench; and repeating a process of forming a conductor layer comprising a via portion that fills the opening for the wiring to form a plurality of conductor layers, wherein the insulating resin layer of the wiring layer is formed by the plurality of the pattern layers, the wiring of the wiring layer is formed by the plurality of the conductor layers, and the trench is formed by connecting the plurality of the openings for the trench.

7. The method according to claim 1, further comprising: repeating a process of forming a pattern layer through exposure and development of a photosensitive resin layer to form a plurality of pattern layers, each of the pattern layers having a pattern including an opening for the wiring; and repeating a process of forming a conductor layer comprising a via portion that fills the opening for the wiring to form a plurality of conductor layers, wherein the insulating resin layer of the wiring layer is formed by the plurality of the pattern layers, the wiring of the wiring layer is formed by the plurality of the conductor layers, and a part of the formed insulating resin layer is removed with laser irradiation to form the trench.
8. The method according to claim 1, further comprising mounting a plurality of semiconductor components on the wiring layer, wherein each of the plurality of wiring structures being formed has one or more of the semiconductor components.
9. The method according to claim 1, wherein the base material further comprises an internal wiring layer exposed on the second main surface, the internal wiring layer forms an internal trench filled with the through portion of the resin portion, the method further comprises mounting a plurality of semiconductor components on the internal wiring layer, the through portion is cut along the trench and the internal trench, and each of the plurality of wiring structures being formed comprises one or more of the semiconductor components.
10. The method according to claim 8, wherein the base material further comprises a relay wiring portion comprising relay wiring electrically connected to the plurality of semiconductor components, the relay wiring portion being sealed by the resin portion, and each of the plurality of wiring structures being formed comprises the relay wiring portion and two or more of the semiconductor components electrically connected via the relay wiring portion.
11. The method according to claim 1, wherein the trench has a width increasing in a direction away from the base material.
12. The method according to claim 1, further comprising mounting the wiring structure on an organic wiring substrate.